

JWT Library Differential Report

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Generated: 2026-09-13T19:38:39.000Z

JWTForge: http://localhost:8787

JWTForge revision: 0.1.4 (9b91cd6fb181dee2ff4ca84df9b81627c35ea20d)

Reference clock: 2026-09-13T19:38:39.000Z; verifier tolerance: 0 seconds.

The policy-unconfigured profile uses each library's key-only verification behavior. For PyJWT and python-jose, non-audience cases supply the known-good audience because JWTForge always emits aud; audience cases receive no audience in this profile.

The configured profile fixes RS256, issuer, and audience through native library options. It also applies a common application-level jku allowlist before library verification; the harness never fetches a header URL.

A divergence is a difference in normalized outcomes (accepted, rejected, or unsupported). It is an investigation lead, not evidence of a vulnerability.

Observe cases deliberately impose no acceptance or rejection requirement and are excluded from required-expectation failure counts.

Flattened and general JWS inputs are classified as unsupported by the harness before a compact-only verifier is called; these rows do not exercise JSON JWS parsers.

Policy-unconfigured profile

Signed control accepted by all libraries: **yes**

Required cases outside their declared expectation: **0**. Observational cases: **9**.

Case	Expected	jose	jsonwebtoken	fast-jwt	PyJWT	Authlib	joserfc	python-jose	Divergent
valid-baseline	accept	accepted	accepted	accepted	accepted	accepted	accepted	accepted	no
scalar-	observe	accepted	accepted	accepted	rejected	accepted	accepted	rejected	yes
audience-									
without-									
expected-									
audience-									
config									

Case	Expected	jose	jsonwebtoken	fast-jwt	PyJWT	Authlib	joserfc	python-jose	Divergent
array-audience-without-expected-audience-config	observe	accepted	accepted	accepted	rejected	accepted	accepted	rejected	yes
unexpected-audience-without-expected-audience-config	observe	accepted	accepted	accepted	rejected	accepted	accepted	rejected	yes
alternate-issuer	observe	accepted	accepted	accepted	accepted	accepted	accepted	accepted	no
expired	reject	rejected	rejected	rejected	rejected	rejected	rejected	rejected	no
future-not-before	reject	rejected	rejected	rejected	rejected	rejected	rejected	rejected	no
null-expiration	reject	rejected	rejected	rejected	rejected	rejected	rejected	rejected	no
string-expiration	observe	rejected	rejected	rejected	accepted	rejected	rejected	accepted	yes
string-issued-at	observe	rejected	accepted	accepted	accepted	rejected	rejected	accepted	yes
numeric-subject	observe	accepted	accepted	accepted	rejected	accepted	rejected	rejected	yes
numeric-jwt-id	observe	accepted	accepted	accepted	rejected	accepted	accepted	rejected	yes
algorithm-none	reject	rejected	rejected	rejected	rejected	rejected	rejected	rejected	no
rsa-signature-with-hs256-header	reject	rejected	rejected	rejected	rejected	rejected	rejected	rejected	no
untrusted-jku-header	observe	accepted	accepted	accepted	accepted	accepted	accepted	accepted	no
trusted-jku-header	accept	accepted	accepted	accepted	accepted	accepted	accepted	accepted	no
signature-removed	reject	rejected	rejected	rejected	rejected	rejected	rejected	rejected	no
signature-replaced	reject	rejected	rejected	rejected	rejected	rejected	rejected	rejected	no
signature-truncated	reject	rejected	rejected	rejected	rejected	rejected	rejected	rejected	no
signature-bit-flip	reject	rejected	rejected	rejected	rejected	rejected	rejected	rejected	no

Case	Expected	jose	jsonwebtoken	fast-jwt	PyJWT	Authlib	joserfc	python-jose	Divergent
payload-tampered	reject	rejected	rejected	rejected	rejected	rejected	rejected	rejected	no
flattened-jws	reject-or-unsupported	unsupported	unsupported	unsupported	unsupported	unsupported	unsupported	unsupported	no
general-jws	reject-or-unsupported	unsupported	unsupported	unsupported	unsupported	unsupported	unsupported	unsupported	no

Configured profile

Signed control accepted by all libraries: **yes**

Required cases outside their declared expectation: **0**. Observational cases: **4**.

Case	Expected	jose	jsonwebtoken	fast-jwt	PyJWT	Authlib	joserfc	python-jose	Divergent
valid-baseline	accept	accepted	accepted	accepted	accepted	accepted	accepted	accepted	no
scalar-audience-with-expected-audience-config	accept	accepted	accepted	accepted	accepted	accepted	accepted	accepted	no
array-audience-with-expected-audience-config	accept	accepted	accepted	accepted	accepted	accepted	accepted	accepted	no
unexpected-audience-with-expected-audience-config	reject	rejected	rejected	rejected	rejected	rejected	rejected	rejected	no
alternate-issuer	reject	rejected	rejected	rejected	rejected	rejected	rejected	rejected	no
expired	reject	rejected	rejected	rejected	rejected	rejected	rejected	rejected	no
future-not-before	reject	rejected	rejected	rejected	rejected	rejected	rejected	rejected	no
null-expiration	reject	rejected	rejected	rejected	rejected	rejected	rejected	rejected	no
string-expiration	observe	rejected	rejected	rejected	accepted	rejected	rejected	accepted	yes

Case	Expected	jose	jsonwebtoken	fast-jwt	PyJWT	Authlib	joserfc	python-jose	Divergent
string-issued-at	observe	rejected	accepted	accepted	accepted	rejected	rejected	accepted	yes
numeric-subject	observe	accepted	accepted	accepted	rejected	accepted	rejected	rejected	yes
numeric-jwt-id	observe	accepted	accepted	accepted	rejected	accepted	accepted	rejected	yes
algorithm-none	reject	rejected	rejected	rejected	rejected	rejected	rejected	rejected	no
rsa-signature-with-hs256-header	reject	rejected	rejected	rejected	rejected	rejected	rejected	rejected	no
untrusted-jku-header	reject	rejected	rejected	rejected	rejected	rejected	rejected	rejected	no
trusted-jku-header	accept	accepted	accepted	accepted	accepted	accepted	accepted	accepted	no
signature-removed	reject	rejected	rejected	rejected	rejected	rejected	rejected	rejected	no
signature-replaced	reject	rejected	rejected	rejected	rejected	rejected	rejected	rejected	no
signature-truncated	reject	rejected	rejected	rejected	rejected	rejected	rejected	rejected	no
signature-bit-flip	reject	rejected	rejected	rejected	rejected	rejected	rejected	rejected	no
payload-tampered	reject	rejected	rejected	rejected	rejected	rejected	rejected	rejected	no
flattened-jws	reject-or-unsupported	unsupported	unsupported	unsupported	unsupported	unsupported	unsupported	unsupported	no
general-jws	reject-or-unsupported	unsupported	unsupported	unsupported	unsupported	unsupported	unsupported	unsupported	no

Case details

- **valid-baseline** (baseline): Compact RS256 signed control with the expected issuer and audience.
- **scalar-audience** (audience): Token contains the expected scalar audience; the policy-unconfigured profile supplies no expected audience.
- **array-audience** (audience): Token contains an audience array including the expected value; the policy-unconfigured profile supplies no expected audience.
- **unexpected-audience** (audience): Token contains an unexpected audience; only the configured profile supplies the expected audience.
- **alternate-issuer** (issuer): Token contains a different issuer; only the configured profile supplies the expected issuer.
- **expired** (time): Expiration is in the past.
- **future-not-before** (time): Not-before time is in the future.
- **null-expiration** (claims): exp is explicitly encoded as JSON null rather than a NumericDate.
- **string-expiration** (claims): exp is encoded as a JSON string rather than a NumericDate.
- **string-issued-at** (claims): iat is encoded as a JSON string rather than a NumericDate.
- **numeric-subject** (claims): sub is encoded as a JSON number rather than a string.

- **numeric-jwt-id** (claims): jti is encoded as a JSON number rather than a string.
- **algorithm-none** (algorithm): Unsigned token declares alg=none.
- **rsa-signature-with-hs256-header** (algorithm): Header declares HS256 while JWTForge signs with the RSA key.
- **untrusted-jku-header** (key-reference): Correctly signed token includes a jku outside the configured allowlist; no adapter fetches it.
- **trusted-jku-header** (key-reference): Correctly signed token includes the configured JWTForge JWKS URL in jku.
- **signature-removed** (signature): Signature is removed from the compact token.
- **signature-replaced** (signature): Signature bytes are replaced.
- **signature-truncated** (signature): Signature is truncated to 16 bytes.
- **signature-bit-flip** (signature): One bit in the signature is flipped.
- **payload-tampered** (payload): Subject is changed without re-signing.
- **flattened-jws** (serialization): Valid signing segments use JWS Flattened JSON Serialization.
- **general-jws** (serialization): Valid signing segments use JWS General JSON Serialization.

Rejection details

Profile	Case	Library	Outcome	Verification attempted	Error classification	Error class	Message
configured	unexpected-audience-with-expected-audience-config	jose	rejected	true	documented-validation-exception	ERR_JWT_CLAIM_UNEXPECTED_AUDIENCE	unexpected claim value
configured	unexpected-audience-with-expected-audience-config	jsonwebtoken	rejected	true	documented-validation-exception	JsonWebTokenError	jwt audience invalid. expected: https://library-differential.example
configured	unexpected-audience-with-expected-audience-config	fast-jwt	rejected	true	documented-validation-exception	FAST_JWT_INVALID_CLAIM_VALUE	The CLAIM value is not allowed.
configured	alternate-issuer	jose	rejected	true	documented-validation-exception	ERR_JWT_CLAIM_UNEXPECTED_ISSUER	unexpected claim value
configured	alternate-issuer	jsonwebtoken	rejected	true	documented-validation-exception	JsonWebTokenError	jwt issuer invalid. expected: http://localhost:8787
configured	alternate-issuer	fast-jwt	rejected	true	documented-validation-exception	FAST_JWT_INVALID_CLAIM_VALUE	The CLAIM value is not allowed.
policy-unconfigured	expired	jose	rejected	true	documented-validation-exception	ERR_JWT_EXPIRED	"Exp" claim timestamp check failed
configured	expired	jose	rejected	true	documented-validation-exception	ERR_JWT_EXPIRED	"Exp" claim timestamp check failed

Profile	Case	Library	Outcome	Verification attempted	Error classification	Error class	Message
policy-unconfigured	expired	jsonwebtoken	rejected	true	documented-validation-exception	TokenExpiredError	jwt expired
configured	expired	jsonwebtoken	rejected	true	documented-validation-exception	TokenExpiredError	jwt expired
policy-unconfigured	expired	fast-jwt	rejected	true	documented-validation-exception	FAST_JWT_EXPIRED	The token has expired at 2026-09-13T19:37:39.000Z.
configured	expired	fast-jwt	rejected	true	documented-validation-exception	FAST_JWT_EXPIRED	The token has expired at 2026-09-13T19:37:39.000Z.
policy-unconfigured	future-not-before	jose	rejected	true	documented-validation-exception	ERR_JWT_CLAIM_VALIDATION_FAILED	timestamp check failed
configured	future-not-before	jose	rejected	true	documented-validation-exception	ERR_JWT_CLAIM_VALIDATION_FAILED	timestamp check failed
policy-unconfigured	future-not-before	jsonwebtoken	rejected	true	documented-validation-exception	NotBeforeError	jwt not active
configured	future-not-before	jsonwebtoken	rejected	true	documented-validation-exception	NotBeforeError	jwt not active
policy-unconfigured	future-not-before	fast-jwt	rejected	true	documented-validation-exception	FAST_JWT_INACTIVE	The token will be active at 2026-09-13T20:38:39.000Z.
configured	future-not-before	fast-jwt	rejected	true	documented-validation-exception	FAST_JWT_INACTIVE	The token will be active at 2026-09-13T20:38:39.000Z.
policy-unconfigured	null-expiration	jose	rejected	true	documented-validation-exception	ERR_JWT_CLAIM_VALIDATION_FAILED	exp must be a number
configured	null-expiration	jose	rejected	true	documented-validation-exception	ERR_JWT_CLAIM_VALIDATION_FAILED	exp must be a number
policy-unconfigured	null-expiration	jsonwebtoken	rejected	true	documented-validation-exception	JsonWebTokenError	invalid exp value
configured	null-expiration	jsonwebtoken	rejected	true	documented-validation-exception	JsonWebTokenError	invalid exp value

Profile	Case	Library	Outcome	Verification attempted	Error classification	Error class	Message
policy-unconfigured	null-expiration	fast-jwt	rejected	true	documented-validation-exception	FAST_JWT_INVALID_CLAIM_TYPE	The ClaimType must be a number.
configured	null-expiration	fast-jwt	rejected	true	documented-validation-exception	FAST_JWT_INVALID_CLAIM_TYPE	The ClaimType must be a number.
policy-unconfigured	string-expiration	jose	rejected	true	documented-validation-exception	ERR_JWT_CLAIM_VALIDATION_FAILED	“expiration” be a number
configured	string-expiration	jose	rejected	true	documented-validation-exception	ERR_JWT_CLAIM_VALIDATION_FAILED	“expiration” be a number
policy-unconfigured	string-expiration	jsonwebtoken	rejected	true	documented-validation-exception	JsonWebTokenError	invalid exp value
configured	string-expiration	jsonwebtoken	rejected	true	documented-validation-exception	JsonWebTokenError	invalid exp value
policy-unconfigured	string-expiration	fast-jwt	rejected	true	documented-validation-exception	FAST_JWT_INVALID_CLAIM_TYPE	The ClaimType must be a number.
configured	string-expiration	fast-jwt	rejected	true	documented-validation-exception	FAST_JWT_INVALID_CLAIM_TYPE	The ClaimType must be a number.
policy-unconfigured	string-issued-at	jose	rejected	true	documented-validation-exception	ERR_JWT_CLAIM_VALIDATION_FAILED	“issuedAt” be a number
configured	string-issued-at	jose	rejected	true	documented-validation-exception	ERR_JWT_CLAIM_VALIDATION_FAILED	“issuedAt” be a number
policy-unconfigured	algorithm-none	jose	rejected	true	documented-validation-exception	ERR_JOSE_NOT_SUPPORTED	Signature not supported either by JOSE or your javascript runtime
configured	algorithm-none	jose	rejected	true	documented-validation-exception	ERR_JOSE_ALG_NOT_ALLOWED	“alg” (Algorithm) Header Parameter value not allowed
policy-unconfigured	algorithm-none	jsonwebtoken	rejected	true	documented-validation-exception	JsonWebTokenError	jwt signature is required
configured	algorithm-none	jsonwebtoken	rejected	true	documented-validation-exception	JsonWebTokenError	jwt signature is required

Profile	Case	Library	Outcome	Verification attempted	Error classification	Error class	Message
policy-unconfigured	algorithm-none	fast-jwt	rejected	true	documented-validation-exception	FAST_JWT_MISSING_SIGNATURE	The JWT signature is missing.
configured	algorithm-none	fast-jwt	rejected	true	documented-validation-exception	FAST_JWT_MISSING_SIGNATURE	The JWT signature is missing.
policy-unconfigured	rsa-signature-with-hs256-header	jose	rejected	true	other-exception	TypeError	CryptoKey instances must be of type “secret” for the HS256 algorithm
configured	rsa-signature-with-hs256-header	jose	rejected	true	documented-validation-exception	ERR_JOSE_ALG_NOT_ALLOWED	Header Parameter value not allowed
policy-unconfigured	rsa-signature-with-hs256-header	jsonwebtoken	rejected	true	documented-validation-exception	JsonWebTokenError	invalid algorithm
configured	rsa-signature-with-hs256-header	jsonwebtoken	rejected	true	documented-validation-exception	JsonWebTokenError	invalid algorithm
policy-unconfigured	rsa-signature-with-hs256-header	fast-jwt	rejected	true	documented-validation-exception	FAST_JWT_INVALID_ALGORITHM	The algorithm is invalid.
configured	rsa-signature-with-hs256-header	fast-jwt	rejected	true	documented-validation-exception	FAST_JWT_INVALID_ALGORITHM	The algorithm is invalid.
configured	untrusted-jku-header	jose	rejected	true	documented-validation-exception	ConfiguredJkuPolicyError	It is not allowlisted: https://attacker.example.com/known/jwks.json
configured	untrusted-jku-header	jsonwebtoken	rejected	true	documented-validation-exception	ConfiguredJkuPolicyError	It is not allowlisted: https://attacker.example.com/known/jwks.json
configured	untrusted-jku-header	fast-jwt	rejected	true	documented-validation-exception	ConfiguredJkuPolicyError	It is not allowlisted: https://attacker.example.com/known/jwks.json
policy-unconfigured	signature-removed	jose	rejected	true	documented-validation-exception	ERR_JWS_SIGNATURE_VERIFICATION_FAILED	signature verification failed

Profile	Case	Library	Outcome	Verification attempted	Error classification	Error class	Message
configured	signature-removed	jose	rejected	true	documented-validation-exception	ERR_JWS_SIGNATURE_VERIFICATION_FAILED	signature verification failed
policy-unconfigured	signature-removed	jsonwebtoken	rejected	true	documented-validation-exception	JsonWebTokenError	jwt signature is required
configured	signature-removed	jsonwebtoken	rejected	true	documented-validation-exception	JsonWebTokenError	jwt signature is required
policy-unconfigured	signature-removed	fast-jwt	rejected	true	documented-validation-exception	FAST_JWT_MISSING_SIGNATURE	The SIGNATURE signature is missing.
configured	signature-removed	fast-jwt	rejected	true	documented-validation-exception	FAST_JWT_MISSING_SIGNATURE	The SIGNATURE signature is missing.
policy-unconfigured	signature-replaced	jose	rejected	true	documented-validation-exception	ERR_JWS_SIGNATURE_VERIFICATION_FAILED	signature verification failed
configured	signature-replaced	jose	rejected	true	documented-validation-exception	ERR_JWS_SIGNATURE_VERIFICATION_FAILED	signature verification failed
policy-unconfigured	signature-replaced	jsonwebtoken	rejected	true	documented-validation-exception	JsonWebTokenError	invalid signature
configured	signature-replaced	jsonwebtoken	rejected	true	documented-validation-exception	JsonWebTokenError	invalid signature
policy-unconfigured	signature-replaced	fast-jwt	rejected	true	documented-validation-exception	FAST_JWT_INVALID_SIGNATURE	The SIGNATURE signature is invalid.
configured	signature-replaced	fast-jwt	rejected	true	documented-validation-exception	FAST_JWT_INVALID_SIGNATURE	The SIGNATURE signature is invalid.
policy-unconfigured	signature-truncated	jose	rejected	true	documented-validation-exception	ERR_JWS_SIGNATURE_VERIFICATION_FAILED	signature verification failed
configured	signature-truncated	jose	rejected	true	documented-validation-exception	ERR_JWS_SIGNATURE_VERIFICATION_FAILED	signature verification failed
policy-unconfigured	signature-truncated	jsonwebtoken	rejected	true	documented-validation-exception	JsonWebTokenError	invalid signature

Profile	Case	Library	Outcome	Verification attempted	Error classification	Error class	Message
configured	signature-truncated	jsonwebtoken	rejected	true	documented-validation-exception	JsonWebTokenError	invalid signature
policy-unconfigured	signature-truncated	fast-jwt	rejected	true	documented-validation-exception	FAST_JWT_INVALID_SIGNATURE	The SIGNATURE signature is invalid.
configured	signature-truncated	fast-jwt	rejected	true	documented-validation-exception	FAST_JWT_INVALID_SIGNATURE	The SIGNATURE signature is invalid.
policy-unconfigured	signature-bit-flip	jose	rejected	true	documented-validation-exception	ERR_JWS_SIGNATURE_VERIFICATION_FAILED	Signature verification failed
configured	signature-bit-flip	jose	rejected	true	documented-validation-exception	ERR_JWS_SIGNATURE_VERIFICATION_FAILED	Signature verification failed
policy-unconfigured	signature-bit-flip	jsonwebtoken	rejected	true	documented-validation-exception	JsonWebTokenError	invalid signature
configured	signature-bit-flip	jsonwebtoken	rejected	true	documented-validation-exception	JsonWebTokenError	invalid signature
policy-unconfigured	signature-bit-flip	fast-jwt	rejected	true	documented-validation-exception	FAST_JWT_INVALID_SIGNATURE	The SIGNATURE signature is invalid.
configured	signature-bit-flip	fast-jwt	rejected	true	documented-validation-exception	FAST_JWT_INVALID_SIGNATURE	The SIGNATURE signature is invalid.
policy-unconfigured	payload-tampered	jose	rejected	true	documented-validation-exception	ERR_JWS_SIGNATURE_VERIFICATION_FAILED	Signature verification failed
configured	payload-tampered	jose	rejected	true	documented-validation-exception	ERR_JWS_SIGNATURE_VERIFICATION_FAILED	Signature verification failed
policy-unconfigured	payload-tampered	jsonwebtoken	rejected	true	documented-validation-exception	JsonWebTokenError	invalid signature
configured	payload-tampered	jsonwebtoken	rejected	true	documented-validation-exception	JsonWebTokenError	invalid signature
policy-unconfigured	payload-tampered	fast-jwt	rejected	true	documented-validation-exception	FAST_JWT_INVALID_SIGNATURE	The SIGNATURE signature is invalid.

Profile	Case	Library	Outcome	Verification attempted	Error classification	Error class	Message
configured	payload-tampered	fast-jwt	rejected	true	documented-validation-exception	FAST_JWT_INVALID_SIGNATURE	The signature is invalid.
policy-unconfigured	flattened-jws	jose	unsupported	false	compact-jwt-only-precheck	UnsupportedSerialization	JWT decoder accepts compact serialization only
configured	flattened-jws	jose	unsupported	false	compact-jwt-only-precheck	UnsupportedSerialization	JWT decoder accepts compact serialization only
policy-unconfigured	flattened-jws	jsonwebtoken	unsupported	false	compact-jwt-only-precheck	UnsupportedSerialization	JWT decoder accepts compact serialization only
configured	flattened-jws	jsonwebtoken	unsupported	false	compact-jwt-only-precheck	UnsupportedSerialization	JWT decoder accepts compact serialization only
policy-unconfigured	flattened-jws	fast-jwt	unsupported	false	compact-jwt-only-precheck	UnsupportedSerialization	JWT decoder accepts compact serialization only
configured	flattened-jws	fast-jwt	unsupported	false	compact-jwt-only-precheck	UnsupportedSerialization	JWT decoder accepts compact serialization only
policy-unconfigured	general-jws	jose	unsupported	false	compact-jwt-only-precheck	UnsupportedSerialization	JWT decoder accepts compact serialization only
configured	general-jws	jose	unsupported	false	compact-jwt-only-precheck	UnsupportedSerialization	JWT decoder accepts compact serialization only
policy-unconfigured	general-jws	jsonwebtoken	unsupported	false	compact-jwt-only-precheck	UnsupportedSerialization	JWT decoder accepts compact serialization only
configured	general-jws	jsonwebtoken	unsupported	false	compact-jwt-only-precheck	UnsupportedSerialization	JWT decoder accepts compact serialization only
policy-unconfigured	general-jws	fast-jwt	unsupported	false	compact-jwt-only-precheck	UnsupportedSerialization	JWT decoder accepts compact serialization only
configured	general-jws	fast-jwt	unsupported	false	compact-jwt-only-precheck	UnsupportedSerialization	JWT decoder accepts compact serialization only
policy-unconfigured	scalar-audience-without-expected-audience-config	PyJWT	rejected	true	documented-validation-exception	InvalidAudienceError	Invalid audience

Profile	Case	Library	Outcome	Verification attempted	Error classification	Error class	Message
policy-unconfigured	scalar-audience-without-expected-audience-config	python-jose	rejected	true	documented-validation-exception	JWTClaimsError	Invalid audience
policy-unconfigured	array-audience-without-expected-audience-config	PyJWT	rejected	true	documented-validation-exception	InvalidAudienceError	Invalid audience
policy-unconfigured	array-audience-without-expected-audience-config	python-jose	rejected	true	documented-validation-exception	JWTClaimsError	Invalid audience
policy-unconfigured	unexpected-audience-without-expected-audience-config	PyJWT	rejected	true	documented-validation-exception	InvalidAudienceError	Invalid audience
configured	unexpected-audience-with-expected-audience-config	PyJWT	rejected	true	documented-validation-exception	InvalidAudienceError	Audience doesn't match
configured	unexpected-audience-with-expected-audience-config	Authlib	rejected	true	documented-validation-exception	InvalidClaimError	invalid_claim: Invalid claim 'aud'
configured	unexpected-audience-with-expected-audience-config	joserfc	rejected	true	documented-validation-exception	InvalidClaimError	invalid_claim: Invalid claim: 'aud'
policy-unconfigured	unexpected-audience-without-expected-audience-config	python-jose	rejected	true	documented-validation-exception	JWTClaimsError	Invalid audience
configured	unexpected-audience-with-expected-audience-config	python-jose	rejected	true	documented-validation-exception	JWTClaimsError	Invalid audience
configured	alternate-issuer	PyJWT	rejected	true	documented-validation-exception	InvalidIssuerError	Invalid issuer
configured	alternate-issuer	Authlib	rejected	true	documented-validation-exception	InvalidClaimError	invalid_claim: Invalid claim 'iss'
configured	alternate-issuer	joserfc	rejected	true	documented-validation-exception	InvalidClaimError	invalid_claim: Invalid claim: 'iss'

Profile	Case	Library	Outcome	Verification attempted	Error classification	Error class	Message
configured	alternate-issuer	python-jose	rejected	true	documented-validation-exception	JWTClaimsError	Invalid issuer
policy-unconfigured	expired	PyJWT	rejected	true	documented-validation-exception	ExpiredSignatureError	Signature has expired
configured	expired	PyJWT	rejected	true	documented-validation-exception	ExpiredSignatureError	Signature has expired
policy-unconfigured	expired	Authlib	rejected	true	documented-validation-exception	ExpiredTokenError	expired_token: The token is expired
configured	expired	Authlib	rejected	true	documented-validation-exception	ExpiredTokenError	expired_token: The token is expired
policy-unconfigured	expired	joserfc	rejected	true	documented-validation-exception	ExpiredTokenError	expired_token: The token is expired
configured	expired	joserfc	rejected	true	documented-validation-exception	ExpiredTokenError	expired_token: The token is expired
policy-unconfigured	expired	python-jose	rejected	true	documented-validation-exception	ExpiredSignatureError	Signature has expired.
configured	expired	python-jose	rejected	true	documented-validation-exception	ExpiredSignatureError	Signature has expired.
policy-unconfigured	future-not-before	PyJWT	rejected	true	documented-validation-exception	ImmatureSignatureError	The token is not yet valid (nbf)
configured	future-not-before	PyJWT	rejected	true	documented-validation-exception	ImmatureSignatureError	The token is not yet valid (nbf)
policy-unconfigured	future-not-before	Authlib	rejected	true	documented-validation-exception	InvalidTokenError	invalid_token: The token is not valid yet
configured	future-not-before	Authlib	rejected	true	documented-validation-exception	InvalidTokenError	invalid_token: The token is not valid yet
policy-unconfigured	future-not-before	joserfc	rejected	true	documented-validation-exception	InvalidClaimError	invalid_claim: The token is not yet valid

Profile	Case	Library	Outcome	Verification attempted	Error classification	Error class	Message
configured	future-not-before	joserfc	rejected	true	documented-validation-exception	InvalidClaimError	invalid_claim: The token is not yet valid
policy-unconfigured	future-not-before	python-jose	rejected	true	documented-validation-exception	JWTClaimsError	The token is not yet valid (nbf)
configured	future-not-before	python-jose	rejected	true	documented-validation-exception	JWTClaimsError	The token is not yet valid (nbf)
policy-unconfigured	null-expiration	PyJWT	rejected	true	other-exception	TypeError	int() argument must be a string, a bytes-like object or a real number, not 'NoneType'
configured	null-expiration	PyJWT	rejected	true	other-exception	TypeError	int() argument must be a string, a bytes-like object or a real number, not 'NoneType'
policy-unconfigured	null-expiration	Authlib	rejected	true	documented-validation-exception	InvalidClaimError	invalid_claim: Invalid claim 'exp'
configured	null-expiration	Authlib	rejected	true	documented-validation-exception	InvalidClaimError	invalid_claim: Invalid claim 'exp'
policy-unconfigured	null-expiration	joserfc	rejected	true	documented-validation-exception	InvalidClaimError	invalid_claim: Claim 'exp' must be a NumericDate value
configured	null-expiration	joserfc	rejected	true	documented-validation-exception	InvalidClaimError	invalid_claim: Claim 'exp' must be a NumericDate value
policy-unconfigured	null-expiration	python-jose	rejected	true	other-exception	TypeError	int() argument must be a string, a bytes-like object or a real number, not 'NoneType'
configured	null-expiration	python-jose	rejected	true	other-exception	TypeError	int() argument must be a string, a bytes-like object or a real number, not 'NoneType'

Profile	Case	Library	Outcome	Verification attempted	Error classification	Error class	Message
policy-unconfigured	string-expiration	Authlib	rejected	true	documented-validation-exception	InvalidClaimError	invalid_claim: Invalid claim 'exp'
configured	string-expiration	Authlib	rejected	true	documented-validation-exception	InvalidClaimError	invalid_claim: Invalid claim 'exp'
policy-unconfigured	string-expiration	joserfc	rejected	true	documented-validation-exception	InvalidClaimError	invalid_claim: Claim 'exp' must be a NumericDate value
configured	string-expiration	joserfc	rejected	true	documented-validation-exception	InvalidClaimError	invalid_claim: Claim 'exp' must be a NumericDate value
policy-unconfigured	string-issued-at	Authlib	rejected	true	documented-validation-exception	InvalidClaimError	invalid_claim: Invalid claim 'iat'
configured	string-issued-at	Authlib	rejected	true	documented-validation-exception	InvalidClaimError	invalid_claim: Invalid claim 'iat'
policy-unconfigured	string-issued-at	joserfc	rejected	true	documented-validation-exception	InvalidClaimError	invalid_claim: Claim 'iat' must be a NumericDate value
configured	string-issued-at	joserfc	rejected	true	documented-validation-exception	InvalidClaimError	invalid_claim: Claim 'iat' must be a NumericDate value
policy-unconfigured	numeric-subject	PyJWT	rejected	true	documented-validation-exception	InvalidSubjectError	Subject must be a string
configured	numeric-subject	PyJWT	rejected	true	documented-validation-exception	InvalidSubjectError	Subject must be a string
policy-unconfigured	numeric-subject	joserfc	rejected	true	documented-validation-exception	InvalidClaimError	invalid_claim: Claim 'sub' must be a StringOrURI value
configured	numeric-subject	joserfc	rejected	true	documented-validation-exception	InvalidClaimError	invalid_claim: Claim 'sub' must be a StringOrURI value

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policy-unconfigured	numeric-subject	python-jose	rejected	true	documented-validation-exception	JWTClaimsError	Subject must be a string.
configured	numeric-subject	python-jose	rejected	true	documented-validation-exception	JWTClaimsError	Subject must be a string.
policy-unconfigured	numeric-jwt-id	PyJWT	rejected	true	documented-validation-exception	InvalidJTIError	JWT ID must be a string
configured	numeric-jwt-id	PyJWT	rejected	true	documented-validation-exception	InvalidJTIError	JWT ID must be a string
policy-unconfigured	numeric-jwt-id	python-jose	rejected	true	documented-validation-exception	JWTClaimsError	JWT ID must be a string.
configured	numeric-jwt-id	python-jose	rejected	true	documented-validation-exception	JWTClaimsError	JWT ID must be a string.
policy-unconfigured	algorithm-none	PyJWT	rejected	true	documented-validation-exception	InvalidAlgorithmError	The specified alg value is not allowed
configured	algorithm-none	PyJWT	rejected	true	documented-validation-exception	InvalidAlgorithmError	The specified alg value is not allowed
policy-unconfigured	algorithm-none	Authlib	rejected	true	documented-validation-exception	UnsupportedAlgorithmError	Unsupported _algorithm:
configured	algorithm-none	Authlib	rejected	true	documented-validation-exception	UnsupportedAlgorithmError	Unsupported _algorithm:
policy-unconfigured	algorithm-none	joserfc	rejected	true	documented-validation-exception	UnsupportedAlgorithmError	Unsupported _algorithm: Algorithm of 'none' is not recommended
configured	algorithm-none	joserfc	rejected	true	documented-validation-exception	UnsupportedAlgorithmError	Unsupported _algorithm: Algorithm of 'none' is not allowed
policy-unconfigured	algorithm-none	python-jose	rejected	true	documented-validation-exception	JWKError	Unable to find an algorithm for key

Profile	Case	Library	Outcome	Verification attempted	Error classification	Error class	Message
configured	algorithm-none	python-jose	rejected	true	documented-validation-exception	JWTError	The specified alg value is not allowed
policy-unconfigured	rsa-signature-with-hs256-header	PyJWT	rejected	true	documented-validation-exception	InvalidAlgorithmError	The specified alg value is not allowed
configured	rsa-signature-with-hs256-header	PyJWT	rejected	true	documented-validation-exception	InvalidAlgorithmError	The specified alg value is not allowed
policy-unconfigured	rsa-signature-with-hs256-header	Authlib	rejected	true	other-exception	KeyError	0
configured	rsa-signature-with-hs256-header	Authlib	rejected	true	documented-validation-exception	UnsupportedAlgorithmError	Unsupported _algorithm:
policy-unconfigured	rsa-signature-with-hs256-header	joserfc	rejected	true	documented-validation-exception	InvalidKeyTypeError	invalid _key_type: Algorithm 'HS256' requires 'oct' key
configured	rsa-signature-with-hs256-header	joserfc	rejected	true	documented-validation-exception	UnsupportedAlgorithmError	Unsupported _algorithm: Algorithm of 'HS256' is not allowed
policy-unconfigured	rsa-signature-with-hs256-header	python-jose	rejected	true	documented-validation-exception	JWKError	Incorrect key type. Expected: 'oct', Received: RSA
configured	rsa-signature-with-hs256-header	python-jose	rejected	true	documented-validation-exception	JWTError	The specified alg value is not allowed
configured	untrusted-jku-header	PyJWT	rejected	true	documented-validation-exception	ConfiguredJkuPolicyError	It is not allowlisted: https://attacker.example.com/known/jwks.json
configured	untrusted-jku-header	Authlib	rejected	true	documented-validation-exception	ConfiguredJkuPolicyError	It is not allowlisted: https://attacker.example.com/known/jwks.json
configured	untrusted-jku-header	joserfc	rejected	true	documented-validation-exception	ConfiguredJkuPolicyError	It is not allowlisted: https://attacker.example.com/known/jwks.json
configured	untrusted-jku-header	python-jose	rejected	true	documented-validation-exception	ConfiguredJkuPolicyError	It is not allowlisted: https://attacker.example.com/known/jwks.json

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policy-unconfigured	signature-removed	PyJWT	rejected	true	documented-validation-exception	InvalidSignatureError	Signature verification failed
configured	signature-removed	PyJWT	rejected	true	documented-validation-exception	InvalidSignatureError	Signature verification failed
policy-unconfigured	signature-removed	Authlib	rejected	true	documented-validation-exception	BadSignatureError	bad_signature:
configured	signature-removed	Authlib	rejected	true	documented-validation-exception	BadSignatureError	bad_signature:
policy-unconfigured	signature-removed	joserfc	rejected	true	documented-validation-exception	BadSignatureError	bad_signature:
configured	signature-removed	joserfc	rejected	true	documented-validation-exception	BadSignatureError	bad_signature:
policy-unconfigured	signature-removed	python-jose	rejected	true	documented-validation-exception	JWTError	Signature verification failed.
configured	signature-removed	python-jose	rejected	true	documented-validation-exception	JWTError	Signature verification failed.
policy-unconfigured	signature-replaced	PyJWT	rejected	true	documented-validation-exception	InvalidSignatureError	Signature verification failed
configured	signature-replaced	PyJWT	rejected	true	documented-validation-exception	InvalidSignatureError	Signature verification failed
policy-unconfigured	signature-replaced	Authlib	rejected	true	documented-validation-exception	BadSignatureError	bad_signature:
configured	signature-replaced	Authlib	rejected	true	documented-validation-exception	BadSignatureError	bad_signature:
policy-unconfigured	signature-replaced	joserfc	rejected	true	documented-validation-exception	BadSignatureError	bad_signature:
configured	signature-replaced	joserfc	rejected	true	documented-validation-exception	BadSignatureError	bad_signature:

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policy-unconfigured	signature-replaced	python-jose	rejected	true	documented-validation-exception	JWTError	Signature verification failed.
configured	signature-replaced	python-jose	rejected	true	documented-validation-exception	JWTError	Signature verification failed.
policy-unconfigured	signature-truncated	PyJWT	rejected	true	documented-validation-exception	InvalidSignatureError	Signature verification failed
configured	signature-truncated	PyJWT	rejected	true	documented-validation-exception	InvalidSignatureError	Signature verification failed
policy-unconfigured	signature-truncated	Authlib	rejected	true	documented-validation-exception	BadSignatureError	bad_signature:
configured	signature-truncated	Authlib	rejected	true	documented-validation-exception	BadSignatureError	bad_signature:
policy-unconfigured	signature-truncated	joserfc	rejected	true	documented-validation-exception	BadSignatureError	bad_signature:
configured	signature-truncated	joserfc	rejected	true	documented-validation-exception	BadSignatureError	bad_signature:
policy-unconfigured	signature-truncated	python-jose	rejected	true	documented-validation-exception	JWTError	Signature verification failed.
configured	signature-truncated	python-jose	rejected	true	documented-validation-exception	JWTError	Signature verification failed.
policy-unconfigured	signature-bit-flip	PyJWT	rejected	true	documented-validation-exception	InvalidSignatureError	Signature verification failed
configured	signature-bit-flip	PyJWT	rejected	true	documented-validation-exception	InvalidSignatureError	Signature verification failed
policy-unconfigured	signature-bit-flip	Authlib	rejected	true	documented-validation-exception	BadSignatureError	bad_signature:
configured	signature-bit-flip	Authlib	rejected	true	documented-validation-exception	BadSignatureError	bad_signature:

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policy-unconfigured	signature-bit-flip	joserfc	rejected	true	documented-validation-exception	BadSignatureError	bad_signature:
configured	signature-bit-flip	joserfc	rejected	true	documented-validation-exception	BadSignatureError	bad_signature:
policy-unconfigured	signature-bit-flip	python-jose	rejected	true	documented-validation-exception	JWTError	Signature verification failed.
configured	signature-bit-flip	python-jose	rejected	true	documented-validation-exception	JWTError	Signature verification failed.
policy-unconfigured	payload-tampered	PyJWT	rejected	true	documented-validation-exception	InvalidSignatureError	Signature verification failed
configured	payload-tampered	PyJWT	rejected	true	documented-validation-exception	InvalidSignatureError	Signature verification failed
policy-unconfigured	payload-tampered	Authlib	rejected	true	documented-validation-exception	BadSignatureError	bad_signature:
configured	payload-tampered	Authlib	rejected	true	documented-validation-exception	BadSignatureError	bad_signature:
policy-unconfigured	payload-tampered	joserfc	rejected	true	documented-validation-exception	BadSignatureError	bad_signature:
configured	payload-tampered	joserfc	rejected	true	documented-validation-exception	BadSignatureError	bad_signature:
policy-unconfigured	payload-tampered	python-jose	rejected	true	documented-validation-exception	JWTError	Signature verification failed.
configured	payload-tampered	python-jose	rejected	true	documented-validation-exception	JWTError	Signature verification failed.
policy-unconfigured	flattened-jws	PyJWT	unsupported	false	compact-jwt-only-precheck	UnsupportedSerializationError	JWT decoder accepts compact serialization only
configured	flattened-jws	PyJWT	unsupported	false	compact-jwt-only-precheck	UnsupportedSerializationError	JWT decoder accepts compact serialization only

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policy-unconfigured	flattened-jws	Authlib	unsupported	false	compact-jwt-only-precheck	UnsupportedSerialization	JWT decoder accepts compact serialization only
configured	flattened-jws	Authlib	unsupported	false	compact-jwt-only-precheck	UnsupportedSerialization	JWT decoder accepts compact serialization only
policy-unconfigured	flattened-jws	joserfc	unsupported	false	compact-jwt-only-precheck	UnsupportedSerialization	JWT decoder accepts compact serialization only
configured	flattened-jws	joserfc	unsupported	false	compact-jwt-only-precheck	UnsupportedSerialization	JWT decoder accepts compact serialization only
policy-unconfigured	flattened-jws	python-jose	unsupported	false	compact-jwt-only-precheck	UnsupportedSerialization	JWT decoder accepts compact serialization only
configured	flattened-jws	python-jose	unsupported	false	compact-jwt-only-precheck	UnsupportedSerialization	JWT decoder accepts compact serialization only
policy-unconfigured	general-jws	PyJWT	unsupported	false	compact-jwt-only-precheck	UnsupportedSerialization	JWT decoder accepts compact serialization only
configured	general-jws	PyJWT	unsupported	false	compact-jwt-only-precheck	UnsupportedSerialization	JWT decoder accepts compact serialization only
policy-unconfigured	general-jws	Authlib	unsupported	false	compact-jwt-only-precheck	UnsupportedSerialization	JWT decoder accepts compact serialization only
configured	general-jws	Authlib	unsupported	false	compact-jwt-only-precheck	UnsupportedSerialization	JWT decoder accepts compact serialization only
policy-unconfigured	general-jws	joserfc	unsupported	false	compact-jwt-only-precheck	UnsupportedSerialization	JWT decoder accepts compact serialization only
configured	general-jws	joserfc	unsupported	false	compact-jwt-only-precheck	UnsupportedSerialization	JWT decoder accepts compact serialization only
policy-unconfigured	general-jws	python-jose	unsupported	false	compact-jwt-only-precheck	UnsupportedSerialization	JWT decoder accepts compact serialization only
configured	general-jws	python-jose	unsupported	false	compact-jwt-only-precheck	UnsupportedSerialization	JWT decoder accepts compact serialization only